

RevoGrocers

Sales Performance Analysis

Sales performance analysis across product categories using SQL (BigQuery)

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RevoU Full Stack Data Analytics | [W4W5 JUN26] | Intermediate

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Business Overview & Disclaimer

Company Overview

- RevoGrocers is a fictional grocery retail business operating across multiple locations and selling a wide range of grocery products.
- It aims to optimize sales strategies, improve customer experience, and maximize revenue through data-driven decisions.

Disclaimer

- This analysis uses a public dataset from Kaggle (Grocery Sales Dataset).
- RevoGrocers is a fictional entity created for educational purposes.
- All insights and recommendations are based on this dataset and do not reflect real-world data.

Dataset Overview & Project Goal

Key Tables

- sales : transactional data
- Products : product details
- categories : product category info

Relevant Fields

- Quantity : units sold
- Price : unit price
- Discount : price discount
- CategoryName : product category
- CustomerID : unique buyers

Project Goal

- Identify the most profitable product categories.
- Analyze the drivers of revenue (units sold vs. customer count).
- Evaluate pricing strategies and their impact on sales.

Entity Relationship Diagram (ERD)

Table relationships in the grocery_dataset that underpin every analysis query.



Methodology

1

Data Setup

Add the grocery_dataset in BigQuery (project fsda-sql-01).

2

Cleaning & Transform

Data cleaning and transformation using SQL in BigQuery.

3

9 Business Questions

Break the analysis down into nine key business questions.

4

Insight (OBIPR)

Frame each query's insight with the OBIPR framework for recommendations.

1 Which category generates the highest revenue after discount?

SQL Query

```
SELECT
  c.categoryname,
  ROUND(SUM(
    s.quantity * p.price * (1 - s.discount)
  ), 2) AS revenue_after_discount
FROM fsda-sql-01.grocery_dataset.sales s
JOIN fsda-sql-01.grocery_dataset.products p
  ON s.productid = p.productid
JOIN fsda-sql-01.grocery_dataset.categories c
  ON p.categoryid = c.categoryid
GROUP BY c.categoryname
ORDER BY revenue_after_discount DESC;
```

Query Output

category	revenue
Confections	556930717.35
Meat	492888844.69
Poultry	440025564.97
Cereals	427393431.91
Snails	372084885.21
Beverages	366515024.01
Produce	362861133.52
Dairy	354358156.77
Seafood	330527987.47
Grain	323879126.65
Shell fish	299598294.03

Insight

Confections generates the highest revenue after discount at 556.93M, ahead of Meat (492.89M) and Poultry (440.03M). This signals strong demand for Confections, making it a top priority for sales strategy.

2

How does revenue after discount relate to total units sold per category?

SQL Query

```
SELECT
  c.categoryname,
  SUM(s.quantity) AS total_units_sold,
  ROUND(SUM(
    s.quantity * p.price * (1 - s.discount)
  ), 2) AS revenue_after_discount
FROM fsda-sql-01.grocery_dataset.sales s
JOIN fsda-sql-01.grocery_dataset.products p
  ON s.productid = p.productid
JOIN fsda-sql-01.grocery_dataset.categories c
  ON p.categoryid = c.categoryid
GROUP BY c.categoryname
ORDER BY revenue_after_discount DESC;
```

Query Output

category	units	revenue
Confections	11078474	556930717.35
Meat	9719292	492888844.69
Poultry	9159847	440025564.97
Cereals	8735296	427393431.91
Snails	7199409	372084885.21
Beverages	7393693	366515024.01
Produce	8174673	362861133.52
Dairy	6815143	354358156.77
Seafood	6996152	330527987.47
Grain	5433152	323879126.65
Shell fish	6983457	299598294.03

Insight

Confections leads with 11.08M units sold and 556.93M revenue. Revenue broadly rises with units sold, but the link is imperfect: Produce sells 8.17M units yet earns less than Cereals (8.74M units) because unit prices differ across categories.

3

How does revenue after discount relate to the number of unique customers per category?

SQL Query

```
SELECT
  c.categoryname,
  COUNT(DISTINCT s.customerid) AS unique_customers,
  ROUND(SUM(
    s.quantity * p.price * (1 - s.discount)
  ), 2) AS revenue_after_discount
FROM fsda-sql-01.grocery_dataset.sales s
JOIN fsda-sql-01.grocery_dataset.products p
  ON s.productid = p.productid
JOIN fsda-sql-01.grocery_dataset.categories c
  ON p.categoryid = c.categoryid
GROUP BY c.categoryname
ORDER BY revenue_after_discount DESC;
```

Query Output

category	customers	revenue
Confections	98743	556930717.35
Meat	98701	492888844.69
Poultry	98679	440025564.97
Cereals	98651	427393431.91
Snails	98376	372084885.21
Beverages	98424	366515024.01
Produce	98601	362861133.52
Dairy	98308	354358156.77
Seafood	98334	330527987.47
Grain	97335	323879126.65
Shell fish	98338	299598294.03

Insight

Unique customers are nearly identical across categories (roughly 97.3k to 98.7k), so revenue is not driven by customer count. The revenue gap comes from purchase volume per customer, not from acquiring more buyers.

4 What is the average price per unit for each category in the catalogue?

SQL Query

```
SELECT
  c.categoryname,
  ROUND(AVG(p.price), 2) AS avg_price_per_unit
FROM fsda-sql-01.grocery_dataset.products p
JOIN fsda-sql-01.grocery_dataset.categories c
  ON p.categoryid = c.categoryid
GROUP BY c.categoryname
ORDER BY avg_price_per_unit DESC;
```

Query Output

category	avg_price
Grain	61.40
Dairy	53.56
Snails	53.20
Meat	52.27
Confections	51.85
Beverages	51.04
Cereals	50.42
Poultry	49.46
Seafood	48.67
Produce	45.80
Shell fish	44.27

Insight

Grain has the highest average price per unit at 61.40, while Shell fish is the lowest at 44.27. This reflects the price positioning of each category in the catalogue.

5

How does average price per unit relate to the number of buyers per category?

SQL Query

```
WITH price_per_category AS (  
    SELECT p.categoryid, c.categoryname,  
           ROUND(AVG(p.price), 2) AS avg_price_per_unit  
    FROM fsda-sql-01.grocery_dataset.products p  
    JOIN fsda-sql-01.grocery_dataset.categories c  
      ON p.categoryid = c.categoryid  
    GROUP BY p.categoryid, c.categoryname  
)  
buyers_per_category AS (  
    SELECT p.categoryid,  
           COUNT(DISTINCT s.customerid) AS unique_buyers  
    FROM fsda-sql-01.grocery_dataset.sales s  
    JOIN fsda-sql-01.grocery_dataset.products p  
      ON s.productid = p.productid  
    GROUP BY p.categoryid  
)  
SELECT pc.categoryname, pc.avg_price_per_unit,  
       bc.unique_buyers  
FROM price_per_category pc  
JOIN buyers_per_category bc  
  ON pc.categoryid = bc.categoryid  
ORDER BY pc.avg_price_per_unit DESC;
```

Query Output

category	avg_price	buyers
Grain	61.40	97335
Dairy	53.56	98308
Snails	53.20	98376
Meat	52.27	98701
Confections	51.85	98743
Beverages	51.04	98424
Cereals	50.42	98651
Poultry	49.46	98679
Seafood	48.67	98334
Produce	45.80	98601
Shell fish	44.27	98338

Insight

The highest-priced category, Grain (61.40), also has the fewest unique buyers (97,335), while lower-priced categories draw slightly more. The spread in buyers is small, so price sensitivity exists but is weak.

6 Which category contributes the most to total revenue (percentage)?

SQL Query

```
WITH category_revenue AS (  
  SELECT c.categoryname,  
         SUM(s.quantity * p.price * (1 - s.discount))  
         AS revenue_after_discount  
  FROM fsda-sql-01.grocery_dataset.sales s  
  JOIN fsda-sql-01.grocery_dataset.products p  
    ON s.productid = p.productid  
  JOIN fsda-sql-01.grocery_dataset.categories c  
    ON p.categoryid = c.categoryid  
  GROUP BY c.categoryname  
)  
SELECT categoryname,  
       ROUND(revenue_after_discount, 2) AS revenue,  
       ROUND(revenue_after_discount * 100.0 /  
             SUM(revenue_after_discount) OVER (), 2)  
       AS revenue_contribution_pct,  
       RANK() OVER (  
         ORDER BY revenue_after_discount DESC  
       ) AS revenue_rank  
FROM category_revenue  
ORDER BY revenue_contribution_pct DESC;
```

Query Output

category	revenue	contrib%	rank
Confections	556930717.35	12.87	1
Meat	492888844.69	11.39	2
Poultry	440025564.97	10.17	3
Cereals	427393431.91	9.88	4
Snails	372084885.21	8.60	5
Beverages	366515024.01	8.47	6
Produce	362861133.52	8.39	7
Dairy	354358156.77	8.19	8
Seafood	330527987.47	7.64	9
Grain	323879126.65	7.48	10
Shell fish	299598294.03	6.92	11

Insight

Confections contributes 12.87% of total revenue after discount, the largest share, followed by Meat (11.39%) and Poultry (10.17%). The top four categories together drive about 44%, while Shell fish contributes the least at 6.92%.

7 Which product category has the highest repeat purchase rate?

SQL Query

```
WITH repeat_customers AS (  
    SELECT c.categoryname, s.customerid  
    FROM fsda-sql-01.grocery_dataset.sales s  
    JOIN fsda-sql-01.grocery_dataset.products p  
        ON s.productid = p.productid  
    JOIN fsda-sql-01.grocery_dataset.categories c  
        ON p.categoryid = c.categoryid  
    GROUP BY c.categoryname, s.customerid  
    HAVING COUNT(DISTINCT s.transactionnumber) > 1  
)  
total_customers AS (  
    SELECT c.categoryname,  
        COUNT(DISTINCT s.customerid) AS total_customers  
    FROM fsda-sql-01.grocery_dataset.sales s  
    JOIN fsda-sql-01.grocery_dataset.products p  
        ON s.productid = p.productid  
    JOIN fsda-sql-01.grocery_dataset.categories c  
        ON p.categoryid = c.categoryid  
    GROUP BY c.categoryname  
)  
SELECT t.categoryname, t.total_customers,  
    COUNT(r.customerid) AS repeat_customers,  
    ROUND(COUNT(r.customerid) * 100.0 /  
        t.total_customers, 2)  
    AS repeat_purchase_rate_pct  
FROM total_customers t  
LEFT JOIN repeat_customers r  
    ON t.categoryname = r.categoryname  
GROUP BY t.categoryname, t.total_customers  
ORDER BY repeat_purchase_rate_pct DESC;
```

Query Output

category	total	repeat	rate%
Confections	98743	98598	99.85
Meat	98701	98318	99.61
Poultry	98679	98122	99.44
Cereals	98651	97867	99.21
Produce	98601	97550	98.93
Beverages	98424	96679	98.23
Snails	98376	96324	97.91
Seafood	98334	96138	97.77
Shell fish	98338	96054	97.68
Dairy	98308	95677	97.32
Grain	97335	91184	93.68

Insight

Confections has the highest repeat purchase rate at 99.85%, with Grain the lowest at 93.68%. Repeat rates are extremely high across all categories (above 93%), indicating strong overall customer loyalty.

8

Summary of all sales performance metrics per category

SQL Query

```
WITH category_metrics AS (
  SELECT c.categoryname,
         SUM(s.quantity) AS total_units_sold,
         COUNT(DISTINCT s.customerid) AS unique_customers,
         SUM(s.quantity * p.price * (1 - s.discount))
           AS revenue_after_discount
  FROM fsda-sql-01.grocery_dataset.sales s
  JOIN fsda-sql-01.grocery_dataset.products p
    ON s.productid = p.productid
  JOIN fsda-sql-01.grocery_dataset.categories c
    ON p.categoryid = c.categoryid
  GROUP BY c.categoryname
)
SELECT categoryname, total_units_sold,
       unique_customers,
       ROUND(revenue_after_discount, 2) AS revenue,
       ROUND(revenue_after_discount * 100.0 /
             SUM(revenue_after_discount) OVER (), 2)
         AS revenue_contribution_pct
FROM category_metrics
ORDER BY revenue DESC;
```

Query Output

category	units	cust.	revenue	%
Confections	11078474	98743	556930717.35	12.87
Meat	9719292	98701	492888844.69	11.39
Poultry	9159847	98679	440025564.97	10.17
Cereals	8735296	98651	427393431.91	9.88
Snails	7199409	98376	372084885.21	8.60
Beverages	7393693	98424	366515024.01	8.47
Produce	8174673	98601	362861133.52	8.39
Dairy	6815143	98308	354358156.77	8.19
Seafood	6996152	98334	330527987.47	7.64
Grain	5433152	97335	323879126.65	7.48
Shell fish	6983457	98338	299598294.03	6.92

Insight

Confections leads on revenue, contribution, and units sold, while customer counts stay uniform (~98k) across categories. Grain commands the highest average price yet posts among the lowest units and revenue, an anomaly worth noting.

9 What is the cumulative transaction of the top user? (window function)

SQL Query

```
WITH user_transactions AS (  
  SELECT s.customerid, s.salesdate,  
         s.transactionnumber,  
         SUM(s.quantity * p.price * (1 - s.discount))  
           AS transaction_value  
  FROM fsda-sql-01.grocery_dataset.sales s  
  JOIN fsda-sql-01.grocery_dataset.products p  
    ON s.productid = p.productid  
  GROUP BY s.customerid, s.salesdate,  
           s.transactionnumber  
)  
,  
top_user AS (  
  SELECT customerid FROM user_transactions  
  GROUP BY customerid  
  ORDER BY SUM(transaction_value) DESC LIMIT 1  
)  
SELECT ut.customerid, ut.salesdate,  
       ut.transactionnumber,  
       ROUND(ut.transaction_value, 2) AS transaction_value,  
       ROUND(SUM(ut.transaction_value) OVER (  
         PARTITION BY ut.customerid  
         ORDER BY ut.salesdate, ut.transactionnumber  
         ROWS BETWEEN UNBOUNDED PRECEDING  
           AND CURRENT ROW), 2) AS cumulative_transaction  
FROM user_transactions ut  
JOIN top_user tu ON ut.customerid = tu.customerid  
ORDER BY ut.salesdate, ut.transactionnumber;
```

Query Output

cust	date	value	cumulative
94800	2018-01-01	2366.35	6104.61
94800	2018-01-04	2251.31	8355.92
94800	2018-01-04	342.45	8698.38
94800	2018-01-05	377.36	9075.74
94800	2018-01-07	1166.35	10242.09
94800	2018-01-08	2245.47	12487.56
94800	2018-01-08	1501.10	13988.66
94800	2018-01-09	1493.92	15482.58
94800	2018-01-10	1111.80	16594.39
...	...	...	...
94800	2018-05-09	1785.78	130324.16

Showing selected rows; running total ends at 130324.16

Insight

The top user is customer 94800, with the highest total transaction value. Their running total climbs steadily, reaching about 130.32k across the observed period, useful for understanding high-value customer behavior.

Summary of Key Findings

- Top revenue-generating category: Confections contributes 12.87% of total revenue, ahead of Meat (11.39%) and Poultry (10.17%).
- Relation between revenue and units sold vs. customer count: Revenue tracks units sold, but customer counts are nearly uniform (~97k to 98.7k), so volume per customer drives revenue, not buyer count.
- Impact of pricing on customer behavior: Price sensitivity is weak. The priciest category, Grain, has the fewest buyers, but the gap across categories is small.
- Category-wise revenue contribution: A few top categories dominate. The top four drive about 44% of total revenue.
- The highest repeat purchase rate is in the Confections category, at 99.85%, with every category above 93%.
- Anomalies or unexpected findings : Grain has the highest average price yet the lowest units sold and near-lowest revenue.

Business Recommendations

Category Prioritization

Focus investment and stock on the highest-revenue categories, led by Confections and Meat.

Pricing Strategy

Review Grain's premium pricing, since its high price pairs with the lowest volume and revenue.

Customer Strategy

Decide whether to focus on acquiring new customers or increasing order size per customer.

Marketing Strategy

Target promotions at high repeat-rate categories like Confections and Meat to lift basket size.

Thank You!!!

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